

- 1 *Smilax china* is a herbaceous plant.

Fig. 1.1 shows part of a transverse section of a root of *S. china* with root hair cells visible.

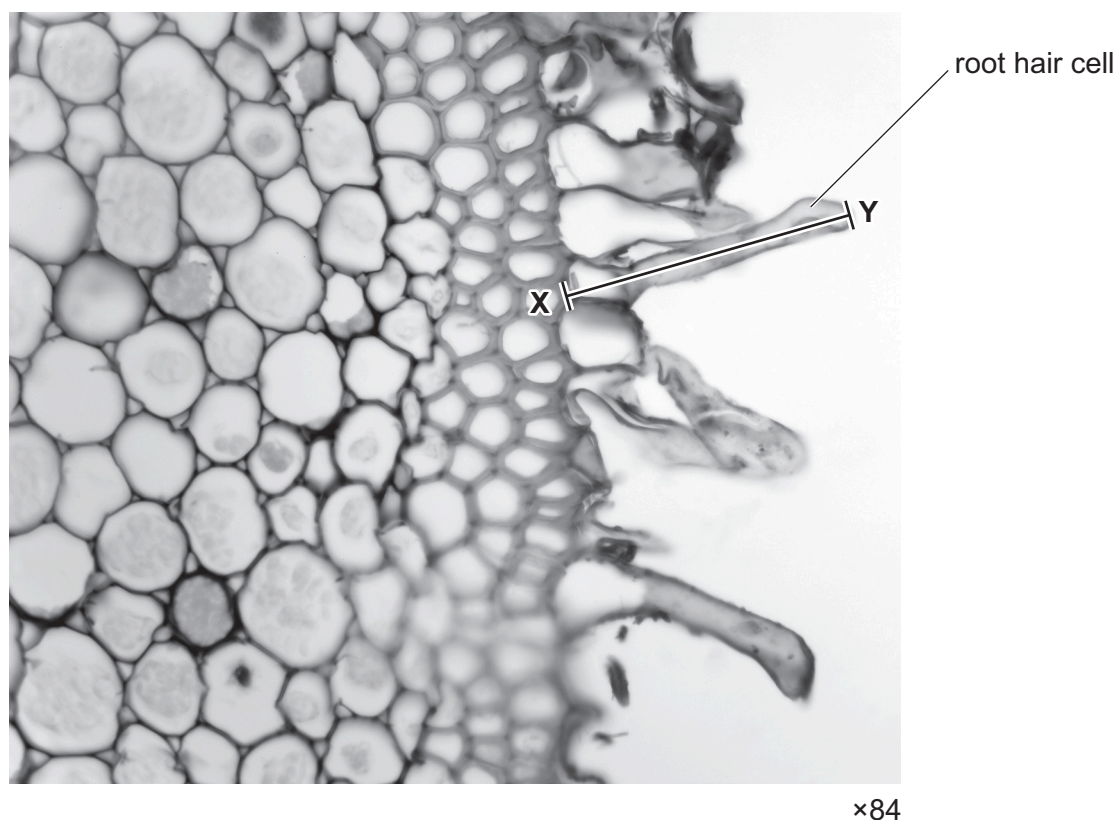


Fig. 1.1

- (a) Name the type of microscope that has been used to obtain the image in Fig. 1.1.

..... [1]

- (b) Calculate the **actual** length, in micrometres (μm), of the root hair cell labelled in Fig. 1.1. Use the **image** length of the root hair cell along line X–Y in your calculation.

actual length = μm [1]

- (c) Root hairs are important adaptations of root hair cells for the uptake of water.

Explain **one** way in which root hairs adapt root hair cells for the uptake of water.

.....
.....
..... [1]

(d) Mineral ions are taken up by root hair cells.

Table 1.1 shows the concentrations of sodium ions (Na^+) and potassium ions (K^+) inside the root hair cells of a plant root and in the soil solution surrounding the root.

Table 1.1

concentration of Na^+ / g dm^{-3}		concentration of K^+ / g dm^{-3}	
root hair cell	soil solution	root hair cell	soil solution
0.35	3.34	5.46	0.16

With reference to Table 1.1:

- suggest the mechanisms involved in the transport of Na^+ and K^+ from the soil solution into the root hair cells
- state the reasons for your suggestions.

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

- (e) The region between the outer layer of a root and the endodermis is known as the cortex. Table 1.2 shows the water potential of two adjacent cells, **A** and **B**, in the cortex of a root.

Table 1.2

water potential / kPa	
cortex cell A	cortex cell B
–120	–350

With reference to Table 1.2, explain the direction of water movement between cell **A** and cell **B**.

.....

.....

.....

.....

..... [2]



- (ii) Dissolved Fe^{2+} is transported across the tissues of the root to the xylem.

When the amount of dissolved Fe^{2+} absorbed by root hair cells is greater than the amount that is needed by the plant, not all of the Fe^{2+} that is absorbed is transported to the xylem.

Suggest how the endodermis can reduce the amount of Fe^{2+} reaching the xylem.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 15]